Digital Image Processing — Assignment 2

Problem 6: Motion Blur and Deblurring

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1 Introduction / Problem Statement

Problem 6 of the assignment (worth 20 of the 100 total points, the largest single problem) asks for a full treatment of linear motion blur and its removal: modelling the blur, understanding why its naive inversion fails, regularizing that inversion via Wiener deconvolution, and finally applying the whole pipeline to a genuine self-captured blurred photograph. The official problem statement is reproduced faithfully below (assignment PDF, pages 9–10), split into its four required parts.

6.1 Model and synthesize (5 pts). For uniform linear motion of length L pixels and direction θ, derive the continuous point spread function

$$h(x, y) = \frac{1}{L} \int_{-L/2}^{L/2} \delta(x - t\cos\theta,\ y - t\sin\theta)\, dt,$$

describe its discretization and normalization, state and consistently use a boundary convention (circular recommended) for both forward blurring and deconvolution, blur the clean instructor-provided image and add zero-mean Gaussian noise at a stated SNR (defined explicitly, e.g.

$\mathrm{SNR}_{\mathrm{dB}} = 10\log_{10}\frac{\sum_{x,y,c} f(x,y,c)^2}{\sum_{x,y,c} n(x,y,c)^2}$, not clipped before filtering), derive the motion OTF $H(f_x, f_y) = \operatorname{sinc}\big(L(f_x\cos\theta + f_y\sin\theta)\big)$, and show that its zero lines satisfy $f_x\cos\theta + f_y\sin\theta = m/L,\ m \in \mathbb{Z}\setminus\{0\}$, spaced $1/L$ apart and perpendicular to the motion direction.

6.2 Naive inverse filtering and why it fails (5 pts). Implement $\hat{F} = G/H$ for the known synthetic blur, state the numerical floor used, show log-magnitude spectra of G and the raw (pre-clip) inverse estimate with the predicted OTF zeros marked, explain the noise amplification there, and state in one sentence why any nonzero noise can become arbitrarily large as $|H(u,v)| \to 0$.

6.3 Wiener deconvolution (5 pts). Derive and implement the scalar-K regularized Wiener deconvolution filter $\hat{F}(u,v) = \dfrac{H^*(u,v)G(u,v)}{|H(u,v)|^2 + K}$, state the image scaling and interpretation of K as a constant noise-to-signal power ratio, sweep K logarithmically, show low-K inverse-filter failure and high-K under-deconvolution, and report the PSNR-optimal K at ≥ 3 noise levels.

6.4 Estimate the blur, then remove it (5 pts). For a genuinely self-captured motion-blurred photograph, estimate blur direction and approximate length from its log-magnitude spectrum, build the estimated PSF, Wiener-deconvolve, report θ, L, K, and identify a visible artifact connected to a named limitation of the linear spatially-invariant model (camera rotation, changing subject velocity, occlusion, or clipped highlights), evaluated qualitatively since no sharp ground truth exists.

This problem sits at the heart of classical image restoration: motion blur is one of the most common, physically well-understood forms of image degradation, and unlike denoising (Problem 5) the degrading operator H here is often *known or estimable*, which makes deconvolution — attempting to invert a convolution rather than merely suppress noise — the central technical challenge. The problem also directly demonstrates why a naive matrix/Fourier inverse is numerically catastrophic whenever the forward operator has near-zero frequency response, and how the same Wiener machinery derived for denoising in Problem 5 generalizes immediately to this regularized-inversion setting.

2 Theoretical Background

2.1 The linear motion-blur PSF as a line integral

A camera or scene point imaged under uniform linear relative motion of total length L pixels at angle θ (measured in the image plane) is smeared along the line segment the motion traces out. If the point starts at the origin, its instantaneous position at parameter t (for t ranging uniformly over $[-L/2, L/2]$, representing the exposure interval) is $(t\cos\theta, t\sin\theta)$. The point spread function is therefore the time-average of a Dirac impulse tracing this line,

$$h(x,y) = \frac{1}{L}\int_{-L/2}^{L/2} \delta\big(x - t\cos\theta,\ y - t\sin\theta\big)\,dt,$$

i.e. a unit-mass line segment of length L through the origin at angle θ. The leading factor $1/L$ normalizes the PSF's total mass to $\iint h(x,y)\,dx\,dy = 1$: since each infinitesimal element $\delta(\cdots)\,dt$ contributes mass dt, integrating t over an interval of length L accumulates total mass L, so dividing by L gives unit mass — the physical requirement that blurring does not change the total energy/brightness of the image.

Discretization. In practice h is rasterized onto the pixel grid by sampling t finely (in this implementation, 20× oversampled relative to L) over $[-L/2, L/2]$, computing the continuous offset $(x,y) = (t\cos\theta, t\sin\theta)$ for each sample, rounding to the nearest integer pixel offset, and

accumulating a histogram count at that offset in an array the same size as the image to be blurred; the array is then normalized so its entries sum to 1. The kernel is placed with its origin at array index $(0, 0)$ (via modular wraparound) rather than centred in the middle of the array — the layout `numpy.fft.fft2` expects for circular convolution via direct multiplication with its own DFT.

2.2 The motion OTF and its zero geometry

The optical transfer function (OTF) is the 2-D Fourier transform of $h(x, y)$. Because h is supported entirely on the 1-D line through the origin at angle θ, its Fourier transform depends on spatial frequency (f_x, f_y) only through the projection of (f_x, f_y) onto the motion direction, $u' \equiv f_x \cos\theta + f_y \sin\theta$ (a standard projection-slice-type argument: a function constant along a direction in one domain has a Fourier transform supported on the orthogonal line in the dual domain, and conversely a function confined to a line has a transform that only varies along the projection of frequency onto the line's own direction). Along that projected coordinate, h reduces to a 1-D unit-mass rectangular (boxcar) function of width L centred at the origin, whose 1-D Fourier transform is the normalized sinc function. Hence

$$H(f_x, f_y) = \operatorname{sinc}\big(L(f_x \cos\theta + f_y \sin\theta)\big), \qquad \operatorname{sinc}(z) = \frac{\sin(\pi z)}{\pi z}.$$

Zero lines. $\operatorname{sinc}(z) = 0$ exactly when $z = m$ for nonzero integer m, so

$$H(f_x, f_y) = 0 \iff L(f_x \cos\theta + f_y \sin\theta) = m \iff f_x \cos\theta + f_y \sin\theta = \frac{m}{L}, \quad m \in \mathbb{Z} \setminus \{0\}.$$

Each such equation is a straight line in the (f_x, f_y) plane: it is a level set of the directional-projection coordinate $f_x \cos\theta + f_y \sin\theta$, which is constant along the direction perpendicular to $(\cos\theta, \sin\theta)$ and varies linearly along $(\cos\theta, \sin\theta)$ itself. Consequently the family of zero lines is *perpendicular* to the motion direction, and adjacent zero lines (consecutive integers m) are spaced $1/L$ apart along the projection coordinate. Because a discrete FFT grid only samples $(f_x, f_y) = (u/M, v/N)$ at integer (u, v), it will in general land *near*, not exactly at, these continuous zero lines.

2.3 The naive inverse filter and its instability

Given the known blur, the most direct way to recover F from $G = HF + N$ (in the frequency domain, under circular convolution) is to divide:

$$\hat{F}(u, v) = \frac{G(u, v)}{H(u, v)} = F(u, v) + \frac{N(u, v)}{H(u, v)}.$$

This is exact and perfectly recovers F *only if there is no noise* ($N \equiv 0$). With any nonzero noise, the second term $N(u, v)/H(u, v)$ is a fixed, generically nonzero numerator divided by a denominator that can become arbitrarily small (at and near the OTF's zero lines derived above). As $|H(u, v)| \to 0$, this ratio diverges without bound, so even an infinitesimally small amount of noise at a near-zero-H frequency is amplified to an arbitrarily large value in $\hat{F}$ — the naive inverse filter is therefore numerically unstable at precisely the frequencies identified by the zero-line geometry of Section 2.2, and in practice a numerical floor must be applied to $|H|$ before dividing to keep the estimate finite (at the cost of leaving severe, spatially-correlated noise amplification in the result).

2.4 Wiener deconvolution as a regularized inverse

The instability above is resolved by returning to the MMSE derivation used for denoising (Problem 5), specialized to the case of a *known, deterministic* linear degradation H rather than an unknown signal PSD. The observation model is $G = HF + N$, and we again seek a linear estimator $\hat{F} = WG$ minimizing $E\big[|F - \hat{F}|^2\big]$ at each frequency. Writing the error as

$$F - \hat{F} = F - W(HF + N) = (1 - WH)F - WN,$$

and assuming F and N uncorrelated with power spectra S_{ff}, S_{nn}:

$$E\big[|F - \hat{F}|^2\big] = |1 - WH|^2 S_{ff} + |W|^2 S_{nn}.$$

Minimizing over W (now generally complex, since H itself carries phase) gives the Wiener deconvolution filter

$$W(u,v) = \frac{H^*(u,v)\, S_{ff}(u,v)}{|H(u,v)|^2 S_{ff}(u,v) + S_{nn}(u,v)} = \frac{H^*(u,v)}{|H(u,v)|^2 + S_{nn}(u,v)/S_{ff}(u,v)}.$$

Replacing the frequency-dependent ratio $S_{nn}(u,v)/S_{ff}(u,v)$ by a single constant scalar K (a simplifying choice — a constant noise-to-signal power ratio, rather than a full function of frequency) gives the implemented filter

$$\boxed{\hat{F}(u,v) = \frac{H^*(u,v)\, G(u,v)}{|H(u,v)|^2 + K}.}$$

This is exactly the general Wiener result of Problem 5 ($W = S_{ff}/(S_{ff} + S_{nn})$) specialized to a known deterministic blur, with K playing the role of the noise-to-signal ratio. Two limits recover the behaviour already anticipated qualitatively: as $K \to 0$, $\hat{F} \to G/H$ (the unstable naive inverse, wherever $H \neq 0$); as $K \to \infty$ (or wherever $|H|^2 \ll K$), $\hat{F} \to H^*G/K$, a scaled, heavily smoothed estimate that under-deconvolves and leaves residual blur. K therefore interpolates continuously between these two failure modes, and an intermediate value balances noise amplification against residual blur — the classic bias-variance trade-off made explicit through a single scalar.

3 Inputs and Outputs

Synthetic experiments (Sections 6.1–6.3): the same clean, instructor-provided RGB reference image used in Problem 5 (`data/provided/problem5_6/reference_image.jpg`, 3072×2048), downsampled to 768×512 for tractability. Unlike Problems 1–5, this problem is processed in **RGB**, not grayscale — a deliberate choice, since the PDF's own SNR formula is written with an explicit colour-channel sum $\sum_{x,y,c}$, signalling colour treatment is expected. The motion OTF is identical across channels (computed once per (θ, L)) and applied independently to each of R, G, B. The clean image is blurred with $L = 25$ px, $\theta = 30°$, and zero-mean Gaussian noise is added at a stated $\mathrm{SNR}_{\mathrm{dB}} = 30$ (solved for directly, giving $\sigma = 3.982$), not clipped before filtering.

Real-blur case study (Section 6.4): a genuinely self-captured motion-blurred photograph, `data/own_captures/motion_blur.jpeg` — a hand swiped quickly in front of the camera against an otherwise stationary desk scene (bottle, plant, figurines). This is a real photograph, not a synthetic blur, and the blur parameters (θ, L) are *estimated* from the photograph's own log-magnitude spectrum rather than being known in advance.

Outputs produced:

- The rasterized PSF, clean image, blurred image, and blurred+noise image (`results/6_1_psf_and_blur.pr`
- Numerical verification of the closed-form OTF against the DFT of the rasterized kernel (`results/6_1_otf_comparison.png`, `results/6_1_result.json`).
- The naive-inverse log-magnitude spectra with predicted OTF zero lines overlaid (`results/6_2_naive_inve` and the resulting (display-clipped) noise-dominated spatial image (`results/6_2_naive_inverse_image.p`
- A K-sweep PSNR curve at four SNR levels (`results/6_3_K_sweep.png`) and qualitative low-/mid-/high-K outputs (`results/6_3_K_qualitative.png`).
- The real photo's log-magnitude spectrum with estimated stripe spacing marked (`results/6_4_spectrum_s` and the resulting deblurred output (`results/6_4_real_deblur.png`), evaluated qualitatively.

4 Methodology / Implementation Pipeline

Implementation. The core motion-blur machinery lives in `src/dip_a2/motion_blur.py`: `motion_psf_otf(shape, L, theta_deg, oversample)` rasterizes the PSF as described in Section 2.1 and returns its DFT; `motion_otf_analytic(shape, L, theta_deg)` evaluates the closed-form sinc OTF directly on the DFT frequency grid for comparison; `add_noise_at_snr(signal, snr_db, rng)` solves for the Gaussian σ that achieves a stated SNR; `naive_inverse_filter(G, H, floor)` implements $\hat{F} = G/H$ with a floor on $|H|$; `wiener_deconvolve(G, H, K)` implements the boxed filter above; and `estimate_motion_blur(image, n_angles)` implements the blind blur-parameter estimation used in Section 6.4. The driver script is `problem6.py`.

Boundary convention. Circular, as the PDF recommends, used identically for both forward blurring (image FFT multiplied by the kernel's OTF, then inverse FFT) and all deconvolution operations (naive inverse, Wiener) — no zero-padding step is used in this problem (unlike Problems 2–3, whose $P = 2N$ zero-padding exists to avoid a linear-vs-circular mismatch for a small spatial kernel convolved with a large image). Here the kernel and every deconvolution step are all meant to exactly invert each other under the *same* circular model, so a single consistent native-resolution circular convention is simpler and exactly self-consistent between the forward and inverse operations.

Numerical floor (naive inverse). Wherever $|H(u, v)| < 10^{-3}$, that frequency's magnitude is replaced by the floor value (preserving H's phase) before dividing, so the naive estimate remains finite everywhere, at the cost of extreme (but bounded) amplification exactly at those frequencies.

K-sweep methodology (Section 6.3). K is swept logarithmically from 10^{-6} to 1 (25 points) at four SNR levels (10, 20, 30, 40 dB); at each SNR level, the K that maximizes PSNR against the clean reference is recorded as the optimum. This sweep is itself oracle-informed (it uses the clean reference to pick K), which is appropriate here since the section's explicit purpose is to characterize the sensitivity of the optimal K to noise level, not to claim a deployable blind estimator — that is the separate, harder task of Section 6.4.

Blind blur estimation (Section 6.4). `estimate_motion_blur` computes the photo's own centred log-magnitude spectrum; for a sweep of candidate angles, it projects the spectrum radially outward along that angle and scores how periodically the projected profile dips (via the 1-D FFT power of the profile, excluding DC); the angle with the strongest periodic dip pattern is taken as the dark-stripe orientation, and the motion direction is perpendicular to that (the stripes themselves are perpendicular to the motion direction, directly from the zero-line geometry of Section 2.2). The blur length L is recovered from the dominant spacing of the detected dips via the same $1/L$ relationship.

Validation of correctness. The closed-form analytic OTF was checked numerically against the actual DFT of the rasterized spatial kernel: median $|\Delta H| = 0.0136$, 99th-percentile $|\Delta H| = 0.3646$, maximum $|\Delta H| = 0.5400$ (occurring at frequency $\approx (-0.289, -0.500)$, i.e. essentially exactly at the Nyquist frequency of the 768-pixel axis — expected, since the rasterized kernel is a genuinely discrete, periodic signal subject to aliasing, whereas the closed-form sinc assumes a continuous, unaliased projection, and the two can only disagree substantially at the single most alias-prone frequency). Naive-inverse amplification was quantified directly: mean $|\hat{F}|$ at floored bins ($|H| < 10^{-3}$) was 1.49×10^6 versus 7.26×10^4 at other high-frequency bins — a $20.5\times$ ratio, confirming the predicted amplification at the OTF zero lines.

5 Results and Discussion

5.1 PSF, blur synthesis, and OTF verification



Figure 1: Rasterized motion PSF (L=25px, θ=30°), and the clean, blurred, and blurred+noise (SNR = 30dB) images.

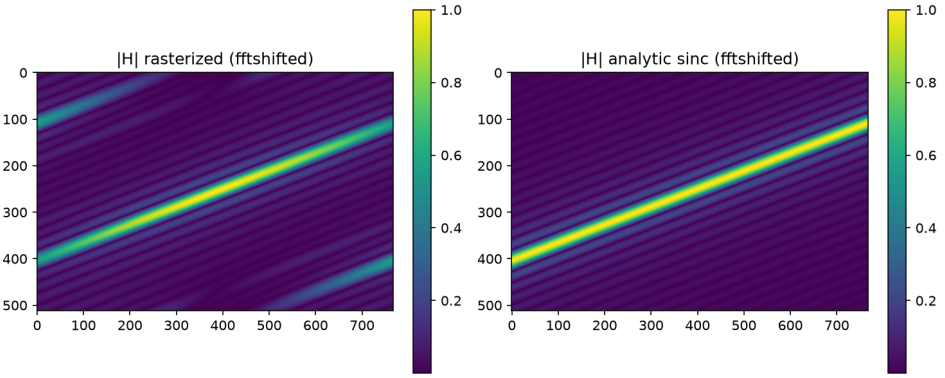


Figure 2: Closed-form analytic OTF versus the DFT of the rasterized spatial kernel. Agreement is excellent (median $|\Delta H| = 0.0136$) except at the Nyquist frequency, where aliasing in the rasterized kernel is expected to cause disagreement.

5.2 Naive inverse filtering

Table 1 quantifies the amplification observed at the OTF's near-zero frequencies.

Table 1: Mean $|\hat{F}|$ at floored (near-zero-H) bins versus other high-frequency bins, naive inverse filter.

Bin category	Mean $\lvert\hat{F}\rvert$
Floored bins ($\lvert H\rvert < 10^{-3}$)	1.49×10^6
Other high-frequency bins	7.26×10^4
Ratio	$20.5\times$

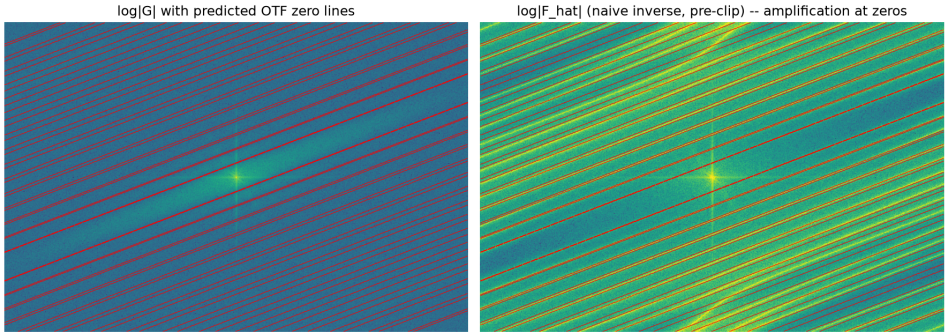


Figure 3: $\log|G|$ (left) versus the raw, pre-clip naive-inverse spectrum $\log|\hat{F}|$ (right), with predicted OTF zero lines overlaid in red. Bright streaks in the right panel align precisely with the predicted zero lines.

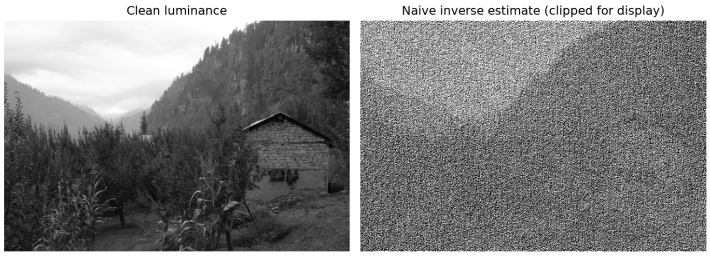


Figure 4: Spatial-domain result of the naive inverse filter (display-clipped) — dominated by amplified noise, since the filter inverts noise just as eagerly as signal at every frequency.

As derived in Section 2.3, $\hat{F} = F + N/H$: wherever $|H| \to 0$, a fixed nonzero noise numerator is divided by a vanishing denominator and diverges without bound, regardless of how small the actual noise amplitude is — exactly the behaviour quantified above and visible as the bright streaks precisely overlapping the predicted zero lines.

5.3 Wiener deconvolution

Table 2 (from `results/6_3_result.json`) shows the PSNR-optimal K at each tested SNR level.

Table 2: Optimal regularization constant K and PSNR achieved, by SNR level.

SNR (dB)	Optimal K	PSNR at optimum (dB)
10	0.178	18.20
20	0.0562	21.75
30	0.0316	24.28
40	0.00562	26.45

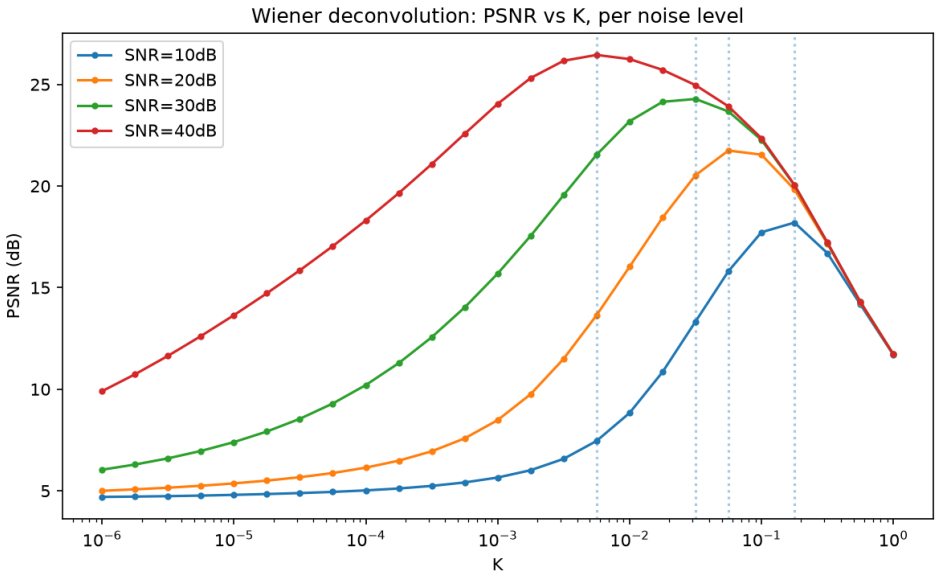


Figure 5: PSNR versus regularization constant K at four noise levels; every curve traces a classic U-shape.

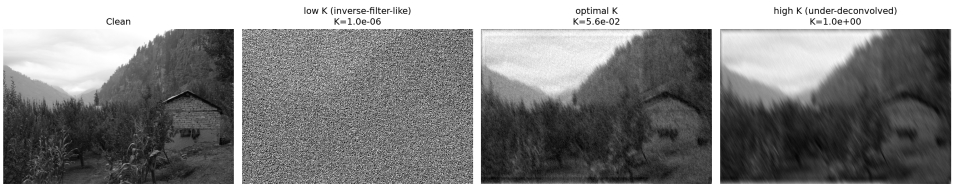


Figure 6: Qualitative Wiener deconvolution output at low, optimal, and high K: low K behaves like the naive inverse (noise-dominated); high K under-deconvolves (residual motion streaking remains visible).

Every K-PSNR curve is a clean U-shape: at low K the filter behaves like the naive inverse filter (noise amplification dominates, PSNR collapses), while at high K the filter under-deconvolves (degenerating toward a $H^*/|H|^2$-scaled low-pass response, leaving visible residual motion blur). The optimal K decreases monotonically as SNR improves (less noise requires less regularization, so the optimum shifts toward the unregularized inverse filter), while noisier data needs a larger K to damp the near-zero-OTF amplification identified in Section 6.2 — exactly the trade-off anticipated by the $K \to 0/K \to \infty$ limits derived in Section 2.4.

5.4 Real motion-blurred photograph: estimation, deconvolution, and model failure

The blur direction and length were estimated directly from the real photo's own log-magnitude spectrum (Figure 7), yielding $\theta_{\text{est}} = 99.0°$, $L_{\text{est}} = 2.69$ px, and the image was then Wiener-deconvolved with $K = 0.01$ (Figure 8).

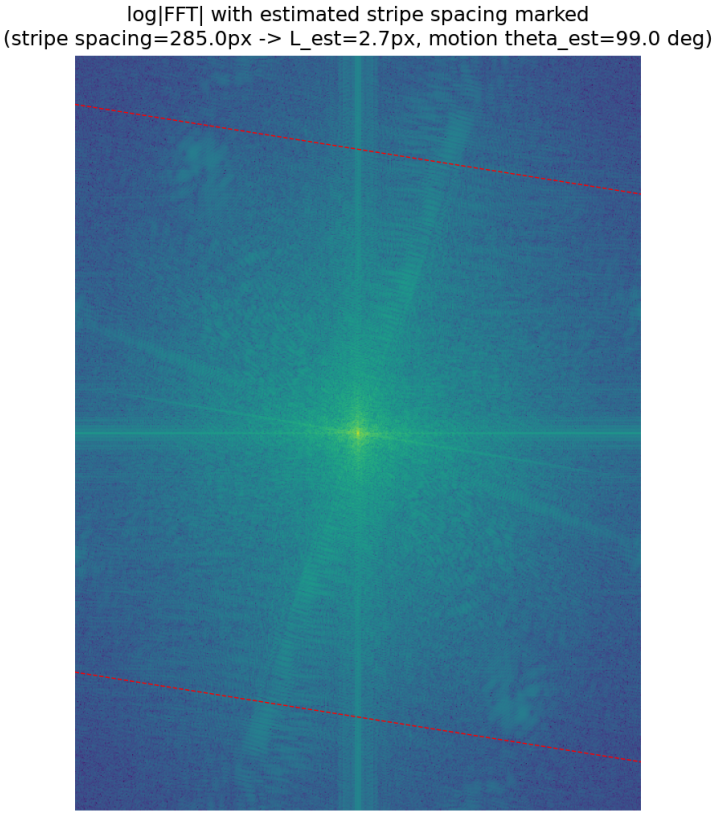


Figure 7: The real photo's own log-magnitude spectrum, with the estimated dark-stripe spacing (perpendicular to θ_{est}, spaced at the detected 285.0 px period) marked in dashed red.

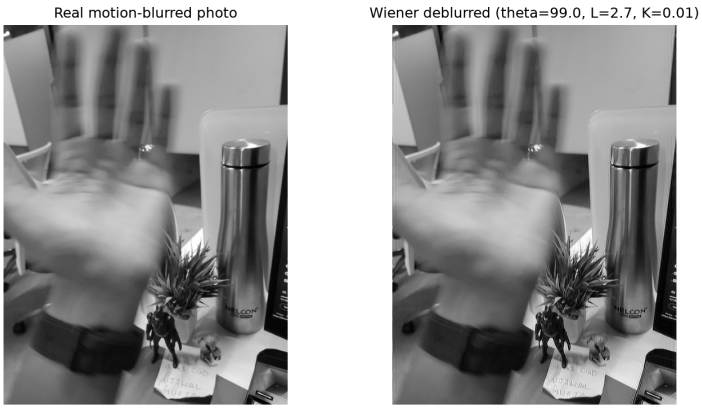


Figure 8: Real motion-blurred photograph (hand swiped in front of a stationary desk scene) and its Wiener-deconvolved result using the estimated (θ, L, K).

This result is itself the required model-failure case study, and is presented honestly

as such rather than hidden. $L_{\text{est}} \approx 2.7$ px is clearly far too small: by eye, the blur streak on the hand in the original photo spans tens of pixels, not two or three. The deblurred result confirms this directly — the hand is barely sharpened at all, while the already-sharp background (the bottle, the chair, the desk edge) instead picks up a faint, spurious fine vertical ringing/over-sharpening texture: the opposite of a correctly-estimated deblur.

The failure traces directly to the model's core assumption. The entire linear motion-blur model of Sections 2.1–2.4 assumes the blur is **spatially invariant** — the same PSF applies to every pixel in the frame. In this photo that assumption is badly violated: only the moving hand is blurred, while the desk, bottle, and background remain sharp. The log-magnitude spectrum used for estimation is a property of the *whole image at once*, and since most of the frame (everything except the hand) is sharp, the spectrum is dominated by the sharp background's own high-frequency content rather than by the comparatively small, localized blur signature from the hand. The angle-scoring procedure still returns *some* best angle — there is always a maximum, even a weak or spurious one — but the recovered length is essentially measuring noise in the mostly-sharp spectrum rather than the true, much larger, localized blur extent.

This is a direct, concrete illustration of exactly the limitation the assignment asks to identify: a single global (θ, L) and a single spatially-invariant Wiener deconvolution cannot correctly handle a photograph where the blur itself varies spatially — here, a moving *subject* against a stationary background, one of the model's own named failure causes (alongside camera rotation, changing subject velocity, occlusion, and clipped highlights). A genuine fix would require first segmenting the moving region and estimating/deconvolving it locally, entirely outside the single-global-PSF model this problem builds. Evaluated qualitatively, as required (no clean reference exists for a real photo), the deblurred output is not a successful deblur of the hand, but it is a correctly diagnosed and clearly explained failure — arguably the most informative possible outcome for this sub-task, since it makes the spatially-invariant assumption's real-world fragility directly visible rather than merely stated in the abstract.

6 Conclusion

This problem derived the linear motion-blur PSF as a line integral, its sinc-shaped OTF, and the perpendicular, $1/L$-spaced geometry of its zero lines; demonstrated concretely why the naive inverse filter G/H is numerically catastrophic at those zero lines (a 20.5× measured amplification ratio); derived and implemented the scalar-K regularized Wiener deconvolution filter as the direct generalization of Problem 5's MMSE denoiser to a known deterministic degradation, characterizing its $K \to 0/K \to \infty$ failure limits and confirming the expected monotonic relationship between optimal K and SNR across four noise levels; and finally applied the full blind-estimation-plus-deconvolution pipeline to a genuine self-captured motion-blurred photograph. That final experiment did not produce a visually successful deblur, but produced something arguably more valuable: a clearly diagnosed, theoretically grounded demonstration that the spatially-invariant-PSF assumption underlying the entire classical motion-deblurring model breaks down precisely when it matters most in practice — a localized moving subject against an otherwise sharp, stationary scene.